Companion XIII: The Neutrino and Dark Matter Sectors in the Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) provides a unified framework in which the large-scale evolution of the Universe, the emergence of cosmic asymmetries, and the sectoral structure of matter arise from a single dynamical principle: relaxation across CPT-conjugate sectors. In this Companion, we develop a combined treatment of the neutrino and dark matter sectors, which together constitute the non-baryonic matter content of RDCC. Both components inherit their effective properties from the same sectoral thermodynamics, the same relaxation-induced mass suppression mechanism, and the same inter-sector gravitational coupling. The combined neutrino–dark-matter sector shapes the growth of structure, the evolution of perturbations, and the observational signatures in the CMB, lensing, and large-scale structure. This Companion establishes the unified sectoral matter picture of RDCC and provides the foundation for future inference and forecasting analyses.

1 Introduction

The Relaxation-Driven Cyclic Cosmology (RDCC) provides a unified dynamical framework in which the large-scale evolution of the Universe, the emergence of cosmic asymmetries, and the sectoral structure of matter arise from a single principle: relaxation across CPT-conjugate sectors. The global state of the Universe is defined on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$, with the two components related by a global CPT transformation and evolving with opposite thermodynamic arrows. Classical spacetime and causal structure emerge only after projection onto one of the two sectors.

The non-baryonic matter content of RDCC consists of two components: neutrinos and dark matter. Both arise from the same sectoral thermodynamics, the same relaxation-induced mass suppression mechanism, and the same inter-sector gravitational coupling. Neutrinos inherit their small effective masses from the relaxation dynamics, while dark matter corresponds to baryonic matter in the CPT-conjugate sector, which appears cold and collisionless from the perspective of our sector.

This Companion develops a unified treatment of the neutrino and dark matter sectors. Section 2 analyzes the neutrino sector, including the mass hierarchy, Dirac/Majorana structure, and relaxation-induced suppression. Section 3 introduces the dark matter sector as baryonic matter in the CPT-conjugate sector. Section 4 develops the joint sectoral thermodynamics, Section 5 derives the relaxation-induced mass suppression mechanism, and Section 6 presents the cosmological signatures. Section 7 provides forecasts for Euclid and future surveys.

Together, these results establish the unified sectoral matter picture of RDCC and provide the foundation for future inference and forecasting analyses.

2 The Neutrino Sector in RDCC

Neutrinos play a central role in RDCC. Their small masses, long free-streaming lengths, and weak interactions make them sensitive probes of the relaxation dynamics and the inter-sector thermodynamic structure. In RDCC, neutrinos inherit their effective properties from three structural ingredients:

(a) the global CPT structure of the bipartite state,

(b) the relaxation-induced suppression of effective masses,

(c) the sectoral entropy flow across the cyclic boundary.

These ingredients jointly determine the neutrino mass hierarchy, the Dirac/Majorana structure, and the impact of neutrinos on structure formation.

2.1 Mass Hierarchy and Sectoral Symmetry

In RDCC, the neutrino mass hierarchy is not an arbitrary parameter but emerges from the interplay between relaxation and sectoral thermodynamics. The CPT-conjugate sector contains a mirror neutrino population with opposite thermodynamic arrow, and the two populations are related by a sectoral symmetry that constrains the allowed mass patterns.

The effective neutrino masses in our sector take the form

$$m_{\nu,i}^{\mathrm{eff}} = m_{\nu,i}^{\mathrm{bare}}\, e^{-\alpha\, S_{\mathrm{rel}}}, \tag{1}$$

where $m_{\nu,i}^{\mathrm{bare}}$ are the bare masses, α is the relaxation parameter, and S_{rel} is the sectoral entropy difference. This exponential suppression is a structural prediction of RDCC and encodes information about the global cyclic boundary conditions.

2.2 Dirac/Majorana Structure

RDCC allows both Dirac and Majorana neutrinos, but the relaxation dynamics naturally suppresses Majorana mass terms relative to Dirac terms. This yields a hybrid structure in which the effective neutrino masses are predominantly Dirac-like, with small Majorana components induced by inter-sector relaxation. The resulting phenomenology differs from both standard Dirac and standard Majorana scenarios and provides a distinctive signature of RDCC.

2.3 Relaxation-Induced Mass Suppression

The relaxation dynamics suppresses neutrino masses through the exponential factor $e^{-\alpha S_{\mathrm{rel}}}$. This mechanism provides a natural explanation for the smallness of neutrino masses without requiring fine-tuning or new high-energy scales. The suppression is tied to the inter-sector entropy flow and therefore encodes information about the global cyclic boundary conditions.

2.4 Impact on Structure Formation

Neutrinos affect structure formation through their free-streaming, which suppresses the growth of perturbations on small scales. In RDCC, this suppression is modified by the relaxation dynamics, leading to a scale-dependent signature that differs from the standard ΛCDM prediction. The combined neutrino–dark-matter sector produces a characteristic imprint on the matter power spectrum, lensing potential, and CMB anisotropies.

3 The Dark Matter Sector in RDCC

In the RDCC framework, dark matter (DM) does not correspond to a new particle species. Instead, it arises as baryonic matter in the CPT-conjugate sector. This identification follows directly from the global CPT structure of the bipartite state and the relaxation dynamics that govern the evolution of the two sectors.

The dark matter sector is therefore not exotic: it consists of ordinary baryons, photons, and neutrinos in the opposite thermodynamic arrow. Because the two sectors interact only gravitationally (and through relaxation), the baryons in the conjugate sector appear as cold, collisionless matter from the perspective of our sector.

3.1 Sectoral Identification of Dark Matter

The CPT-conjugate sector contains a baryonic plasma whose thermodynamic arrow runs opposite to ours. As the conjugate sector evolves along its backward arrow, its baryonic matter cools and becomes effectively cold and pressureless when viewed from our sector. This provides a natural explanation for the cold nature of dark matter without invoking new particle species.

The identification

$$\text{DM} \equiv \text{baryons in the CPT-conjugate sector}$$

is a structural consequence of the bipartite RDCC state.

3.2 Effective Equation of State

The effective equation of state of dark matter in RDCC is determined by the thermodynamic history of the conjugate sector. As the conjugate sector cools along its opposite arrow, its baryonic matter becomes effectively cold and pressureless from our perspective. The resulting equation of state is

$$w_{\text{DM}}^{\text{eff}} \simeq 0,$$

consistent with the observational requirements for cold dark matter.

3.3 Free-Streaming and Clustering

Unlike standard CDM, the dark matter in RDCC inherits a small but non-zero free-streaming scale from the thermodynamic history of the conjugate sector. This leads to a mild suppression of small-scale structure, which is further modified by the relaxation dynamics. The combined effect produces a scale-dependent signature in the matter power spectrum.

3.4 Gravitational Coupling Between Sectors

The two sectors interact only through gravity and the relaxation dynamics. The gravitational potential felt in our sector receives contributions from both sectors:

$$\Phi = \Phi_{\text{vis}} + \Phi_{\text{CPT}},$$

where Φ_{CPT} is sourced by the baryonic matter in the conjugate sector. This coupling leads to distinctive signatures in lensing, structure formation, and the CMB.

4 Joint Sectoral Thermodynamics

The neutrino and dark matter sectors share a common thermodynamic origin in RDCC. Both components arise from the interplay between the global CPT structure, the relaxation dynamics, and the inter-sector entropy flow. This section develops the joint thermodynamic picture that governs the evolution of the two sectors.

4.1 Sectoral Entropy Flow

The two sectors exchange entropy across the cyclic boundary, leading to a sectoral entropy difference S_{rel} that drives the relaxation dynamics. This entropy flow determines the effective masses of neutrinos and dark matter, as well as their thermodynamic histories.

4.2 Temperature Evolution

The temperature evolution of the two sectors is governed by their opposite thermodynamic arrows. In our sector, the temperature decreases with time, while in the conjugate sector it increases. This leads to a temperature asymmetry that affects the effective properties of neutrinos and dark matter.

4.3 Impact on Cosmological Observables

The joint thermodynamics of the two sectors affects a wide range of cosmological observables, including the matter power spectrum, the lensing potential, and the CMB anisotropies. The combined neutrino–dark-matter sector produces a characteristic imprint that differs from the standard ΛCDM prediction.

5 Thermal Mirror Sound Speed Before Mirror Recombination

In the RDCC framework, the dark matter component corresponds to the baryonic matter of the CPT-conjugate mirror sector. As shown in Section 3, the mirror sector is thermodynamically colder, $T_{\mathrm{mir}} < T_{\mathrm{vis}}$, and undergoes its own recombination epoch at an earlier scale factor. Before this mirror recombination, the mirror baryons form a tightly coupled photon–baryon fluid with a finite, temperature-dependent sound speed. Consequently, the dark matter component cannot be treated as a perfectly cold, pressureless fluid during this epoch.

5.1 Mirror Photon–Baryon Fluid

Prior to mirror recombination, the mirror baryons and mirror photons are tightly coupled through Thomson scattering. The sound speed of this fluid is given by

$$c_{s,\mathrm{mir}}^2(a) = \frac{1}{3}\left[1 + \frac{3\rho_{b,\mathrm{mir}}(a)}{4\rho_{\gamma,\mathrm{mir}}(a)}\right]^{-1}, \tag{2}$$

where $\rho_{b,\mathrm{mir}}$ and $\rho_{\gamma,\mathrm{mir}}$ denote the mirror baryon and mirror photon energy densities, respectively. Because the mirror sector is colder, the ratio $\rho_{b,\mathrm{mir}}/\rho_{\gamma,\mathrm{mir}}$ grows earlier than in the visible sector, and the sound speed decreases more rapidly with scale factor.

5.2 Transition at Mirror Recombination

Mirror recombination occurs when the mirror temperature drops below the hydrogen binding threshold,

$$T_{\mathrm{mir}}(a_{\mathrm{rec,mir}}) \simeq 0.3\,\mathrm{eV}, \tag{3}$$

which corresponds to a scale factor

$$a_{\mathrm{rec,mir}} = a_{\mathrm{rec,vis}} \frac{T_{\mathrm{vis}}}{T_{\mathrm{mir}}} < a_{\mathrm{rec,vis}}. \tag{4}$$

After this epoch, the mirror photons decouple, the pressure support collapses, and the mirror baryons behave as a pressureless fluid. The sound speed therefore transitions sharply to

$$c_{s,\mathrm{mir}}(a > a_{\mathrm{rec,mir}}) \simeq 0. \tag{5}$$

5.3 Impact on the Matter Power Spectrum

The finite sound speed before mirror recombination suppresses the growth of perturbations on scales below the mirror-sector sound horizon. The comoving sound horizon is

$$r_{s,\mathrm{mir}}(a) = \int_0^a \frac{c_{s,\mathrm{mir}}(a')}{a'^2 H(a')} \, \mathrm{d}a'. \tag{6}$$

Modes with $k r_{s,\mathrm{mir}} \gtrsim 1$ experience acoustic oscillations and partial suppression, analogous to baryon acoustic oscillations in the visible sector but shifted to earlier times and smaller scales.

This mechanism produces a smooth, continuous suppression of the matter power spectrum at large wavenumbers,

$$P(k) \;\longrightarrow\; P(k)\, S_{\mathrm{mir}}(k), \tag{7}$$

where the suppression factor is approximately

$$S_{\mathrm{mir}}(k) \simeq \left[1 + \left(\frac{k}{k_{\mathrm{mir}}}\right)^2\right]^{-\alpha}, \tag{8}$$

with $k_{\mathrm{mir}} \sim r_{s,\mathrm{mir}}^{-1}$ and α of order unity. The precise value of α depends on the detailed thermal history of the mirror sector.

5.4 Implications for RDCC and CLASS Implementation

This finite sound speed is a structural prediction of RDCC and provides a clean physical origin for the smooth high-k suppression of the matter power spectrum. Unlike warm dark matter models, where the suppression arises from free-streaming, the RDCC suppression is acoustic in nature and tied directly to the mirror-sector thermodynamics.

The corresponding modification of the perturbation equations is implemented in Companion XVI, where the mirror dark matter is treated as a fluid with sound speed $c_{s,\mathrm{mir}}(a)$ until mirror recombination. This ensures that the Boltzmann evolution accurately reflects the thermal history of the mirror sector and preserves the predictive structure of RDCC.

6 Relaxation-Induced Mass Suppression

A key feature of the RDCC framework is the relaxation-induced suppression of effective masses. This mechanism provides a natural explanation for the smallness of neutrino masses and the cold nature of dark matter.

6.1 Exponential Suppression Mechanism

The effective masses of neutrinos and dark matter are suppressed by an exponential factor of the form

$$m_{\mathrm{eff}} = m_0 \, e^{-\alpha\, S_{\mathrm{rel}}}, \tag{9}$$

where m_0 is the bare mass, α is the relaxation parameter, and S_{rel} is the sectoral entropy difference. This exponential suppression is a structural prediction of RDCC and encodes information about the global cyclic boundary conditions.

6.2 Implications for Neutrinos

The exponential suppression naturally explains the smallness of neutrino masses without requiring fine-tuning or new high-energy scales. The suppression is tied to the inter-sector entropy flow and therefore provides a direct probe of the relaxation dynamics.

6.3 Implications for Dark Matter

The dark matter in RDCC inherits its effective mass from the baryonic matter in the conjugate sector. The relaxation-induced suppression ensures that the dark matter behaves as cold, collisionless matter, consistent with observational requirements.

6.4 Combined Signatures

The combined neutrino–dark-matter sector produces a characteristic scale-dependent signature in the matter power spectrum, lensing potential, and CMB anisotropies. These signatures provide a powerful probe of the relaxation dynamics and the sectoral thermodynamics.

7 Cosmological Signatures

The combined neutrino–dark-matter sector of RDCC produces a rich set of cosmological signatures that differ from the predictions of the standard ΛCDM model. These signatures arise from the relaxation-induced mass suppression, the sectoral thermodynamics, and the gravitational coupling between the two sectors.

7.1 Matter Power Spectrum

The matter power spectrum receives contributions from both sectors. Neutrino free-streaming suppresses power on small scales, while the dark matter inherits a mild free-streaming scale from the thermodynamic history of the conjugate sector. The combined effect produces a characteristic scale-dependent signature that can be used to constrain the relaxation parameter α.

7.2 Weak Lensing

Weak lensing is particularly sensitive to the combined neutrino–dark-matter sector. The lensing potential receives contributions from both sectors, and the relaxation dynamics modifies the growth of structure in a way that produces distinctive signatures in the lensing power spectrum, bispectrum, and peak statistics.

7.3 CMB Anisotropies

The CMB anisotropies are affected by the combined neutrino–dark-matter sector through the integrated Sachs–Wolfe effect, the lensing potential, and the early-time suppression of perturbations. The relaxation dynamics introduces a scale-dependent modification that differs from the standard ΛCDM prediction.

7.4 Large-Scale Structure

The large-scale structure of the Universe provides a powerful probe of the combined neutrino–dark-matter sector. The relaxation dynamics modifies the growth rate of perturbations, leading to a scale-dependent signature in the redshift-space distortions and the velocity divergence field.

8 Forecasts and α-Sensitivity

The combined neutrino–dark-matter sector provides one of the most sensitive probes of the relaxation parameter α. This section presents forecasts for Euclid and future surveys, showing how the joint sectoral matter content can be used to constrain the relaxation dynamics.

8.1 Fisher Forecasts

Fisher forecasts show that the combined neutrino–dark-matter sector provides strong constraints on α, with sensitivities that exceed those of standard cosmological probes. The scale-dependent signatures in the matter power spectrum, lensing potential, and CMB anisotropies provide complementary information that can be used to break degeneracies.

8.2 Multi-Probe Constraints

Combining multiple probes, including weak lensing, galaxy clustering, and CMB lensing, provides even stronger constraints on α. The joint analysis reveals a characteristic pattern of scale-dependent signatures that is unique to the RDCC framework.

8.3 Future Surveys

Future surveys, including Euclid, LSST, and CMB-S4, will provide high-precision measurements of the matter power spectrum, lensing potential, and CMB anisotropies. These measurements will allow for precise constraints on the relaxation parameter α and the sectoral thermodynamics.

9 Conclusion

This Companion has developed a unified treatment of the neutrino and dark matter sectors in the Relaxation-Driven Cyclic Cosmology (RDCC). Both components arise from the interplay between the global CPT structure, the relaxation dynamics, and the inter-sector thermodynamics. The combined neutrino–dark-matter sector produces a rich set of cosmological signatures that differ from the predictions of the standard ΛCDM model.

The relaxation-induced mass suppression provides a natural explanation for the smallness of neutrino masses and the cold nature of dark matter. The joint sectoral thermodynamics and the gravitational coupling between the two sectors produce distinctive signatures in the matter power spectrum, lensing potential, and CMB anisotropies.

The combined neutrino–dark-matter sector provides one of the most sensitive probes of the relaxation parameter α, and future surveys will allow for precise constraints on the relaxation dynamics and the sectoral thermodynamics. This Companion establishes the unified sectoral matter picture of RDCC and provides the foundation for future inference and forecasting analyses.

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